

Positive integer solutions of a symmetric generalized Markov equation

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Abstract

We classify the positive integer solutions of $x^2 + y^2 + z^2 + xy + xz + yz = kxyz$ for every positive integer parameter k . Solutions exist exactly for $k \in \{1, 2, 3, 6\}$. For each such parameter, every solution is obtained from $(6/k, 6/k, 6/k)$ by coordinate permutations and Vieta involutions. The proof uses descent in the coordinate sum and an explicit classification of the terminal triples: at a terminal solution with $x \leq y \leq z$, one has $4 \leq kx \leq 6$, and fourteen discriminant calculations exclude $kx = 4, 5$. The case $k = 6$ is known. The argument addresses this fixed symmetric specialization of a broader coefficient-family question and makes no claim of priority.

1 Statement and context

For $k \in \mathbb{Z}_{>0}$, write

$$S_k = \{(x, y, z) \in \mathbb{Z}_{>0}^3 : x^2 + y^2 + z^2 + xy + xz + yz = kxyz\}. \quad (1)$$

The associated Vieta transformations, with k fixed, are

$$\begin{aligned} \mu_x(x, y, z) &= (kyz - x - y - z, y, z), \\ \mu_y(x, y, z) &= (x, kxz - x - y - z, z), \\ \mu_z(x, y, z) &= (x, y, kxy - x - y - z). \end{aligned} \quad (2)$$

An orbit means the set obtained by all finite compositions of these transformations and coordinate permutations, including the empty composition.

Theorem 1. *Let k be a positive integer. The set S_k is nonempty if and only if $k \in \{1, 2, 3, 6\}$. For each of these four values, S_k is exactly the orbit of $(6/k, 6/k, 6/k)$ under coordinate permutations and the three transformations in (2).*

There is no primitivity assumption or coordinate bound. The assertion is exhaustion of the solution set by one seed orbit; it does not assert uniqueness of a word in the transformations.

Chen and Huang [1, Question 7.1] ask for the admissible parameters and solution orbits of

$$x^2 + y^2 + z^2 + k_1yz + k_2xz + k_3xy = kxyz$$

for fixed nonnegative integer coefficients k_1, k_2, k_3 . Equation (1) is the specialization $k_1 = k_2 = k_3 = 1$. Gyoda and Matsushita [2, Theorem 1] enumerate all positive integer solutions of the normalized

family $k = 3 + k_1 + k_2 + k_3$ from $(1, 1, 1)$. Thus their theorem already gives our $k = 6$ case, as recorded in [1, Remark 7.2(1)]. Gyoda and Matsushita also attribute this symmetric case to an earlier theorem of Gyoda [2, p. 3].

Relative to that normalized-family theorem, the assertions addressed here are exclusion of every other positive integer k and exhaustion of the solution sets for $k = 1, 2, 3$. Scaling a known $k = 6$ solution by $6/k$ produces solutions for these three parameters, but does not by itself show that every solution is obtained in this way. The terminal classification below provides the required exclusion and exhaustion. The proof is self-contained. The comparison is limited to the cited Chen–Huang and Gyoda–Matsushita sources; the earlier Gyoda theorem is reported through the latter paper. No comprehensive historical comparison or priority claim is intended, and the general coefficient-family question is outside the scope of this note.

2 Descent and terminal classification

Lemma 1. *The transformations in (2) are involutions of S_k . Every element of S_k can be carried by these involutions and permutations to an ordered solution satisfying*

$$x \leq y \leq z \leq z', \quad z' = kxy - x - y - z. \quad (3)$$

Proof. For a solution (x, y, z) , consider the monic quadratic

$$F(T) = T^2 + (x + y - kxy)T + x^2 + xy + y^2. \quad (4)$$

Its roots are z and

$$z' = kxy - x - y - z = \frac{x^2 + xy + y^2}{z}. \quad (5)$$

The first expression proves integrality and the second strict positivity. Consequently $(x, y, z') \in S_k$, and replacing z' by its companion root returns z . Symmetry gives the other two involutions and preservation under permutations.

Order a solution as $x \leq y \leq z$. Whenever $z' < z$, replace z by z' and reorder. The positive integer height $x + y + z$ strictly decreases, while reordering leaves it unchanged. Thus only finitely many such steps are possible, and the last solution satisfies (3). This argument also applies when maximal coordinates are tied. $\square$

Call an ordered solution satisfying (3) *terminal*.

Lemma 2. *Every terminal solution is diagonal and satisfies $kx = 6$. Hence the full list of terminal solutions, with their parameters, is*

$$(k; x, y, z) = (1; 6, 6, 6), (2; 3, 3, 3), (3; 2, 2, 2), (6; 1, 1, 1). \quad (6)$$

Proof. Put $m = kx$. The root sum and (3) imply

$$2y \leq z + z' = (m - 1)y - x,$$

so $(m - 3)y \geq x > 0$, and therefore the integer m is at least 4. Also $y - z \leq 0$ and $y - z' \leq 0$, whence

$$0 \leq F(y) = (y - z)(y - z') = x^2 + 2xy + (3 - m)y^2 \leq (6 - m)y^2. \quad (7)$$

The last inequality follows from $x \leq y$. Since $y > 0$, we obtain

$$4 \leq m \leq 6. \quad (8)$$

If $m = 6$, then $F(y) = (x - y)(x + 3y) \geq 0$. As $x \leq y$ and $x + 3y > 0$, this forces $x = y$. Substituting in (4) gives

$$F(T) = T^2 - 4xT + 3x^2 = (T - x)(T - 3x).$$

The condition $z \leq z'$ selects the smaller root, so $z = x$. This yields (6).

It remains to exclude $m = 4, 5$. In either case (7) forces $y < 3x$: otherwise

$$F(y) \leq x^2 + 2xy - y^2 \leq \left(\frac{1}{9} + \frac{2}{3} - 1\right)y^2 = -\frac{2}{9}y^2 < 0.$$

Since $kx = m$, there are only three possible factor pairs (k, x) for $m = 4$ and two for $m = 5$. Table 1 lists every integer y satisfying $y \geq x$ and $F(y) \geq 0$ for these pairs, and the corresponding discriminant

$$D = ((m - 1)y - x)^2 - 4(x^2 + xy + y^2). \quad (9)$$

m	k	x	Admissible y	Discriminants, in the same order
4	1	4	4, 5, 6, 7, 8, 9	-128, -123, -108, -83, -48, -3
4	2	2	2, 3, 4	-32, -27, -12
4	4	1	1, 2	-8, -3
5	1	5	5, 6	-75, -3
5	5	1	1	-3

Table 1: All fourteen remaining terminal candidates have negative discriminant.

For completeness, the ranges in the table can be checked without a search program. For fixed m, x , the forward difference of $G(y) = x^2 + 2xy + (3 - m)y^2$ is

$$G(y + 1) - G(y) = 2x + (3 - m)(2y + 1) < 0 \quad (y \geq x).$$

For $m = 4$, the last included and first excluded values give

x	$(y, y + 1)$	$G(y)$	$G(y + 1)$
4	(9, 10)	7	-4
2	(4, 5)	4	-1
1	(2, 3)	1	-2

and for $m = 5$ they give

x	$(y, y + 1)$	$G(y)$	$G(y + 1)$
5	(6, 7)	13	-3
1	(1, 2)	1	-3.

These boundary checks and strict decrease establish that no admissible integer y is omitted. Each discriminant in Table 1 is negative, whereas the roots z, z' of F would give $D = (z - z')^2 \geq 0$. Thus $m = 4, 5$ are impossible, completing the terminal classification. $\square$

Proof of Theorem 1. Every solution descends by Lemma 1, with k unchanged, to a terminal solution. Lemma 2 therefore implies $k \in \{1, 2, 3, 6\}$. Conversely, for each listed k , the positive integer $t = 6/k$ satisfies $6t^2 = kt^3$, so $(t, t, t) \in S_k$.

For any solution at this parameter, reverse its finite descent to the seed. Each mutation is its own inverse, and the inverse of a coordinate permutation is allowed. This gives a finite composition carrying the seed to the chosen solution, with every intermediate triple positive and integral. Conversely, preservation under all the transformations shows that every triple in the seed orbit belongs to S_k . The two sets are equal. $\square$

3 Exact checks and limitations

The finite calculation used by the proof is completely displayed in Table 1 and its boundary checks. The accompanying `verify_table.py` regenerates its fourteen discriminants using integer arithmetic. No unbounded numerical search is needed for the proof.

Supplementary computations during the investigation tested all 20,958,500 ordered triples $1 \leq x \leq y \leq z \leq 500$, deriving k by exact divisibility rather than imposing a parameter cutoff. They found nineteen solutions. A separate implementation recovered the same list by solving the integer quadratics with exact square-root tests. Mutation, descent, and reversal were also checked on these solutions. These are finite consistency checks; the height argument and terminal classification establish the unbounded theorem.

As broad adjacent Diophantine methodology, Zeng’s work on regularity obstructions and certified searches for rational Diophantine septuples [3] likewise studies algebraic transformations alongside exact arithmetic. It is cited for that general methodological context and supplies no lemma used in this proof.

Computational and AI assistance. OpenAI Codex assisted with proof development, exact-integer verification programs, source comparison, a separate mathematical review pass, and manuscript preparation. The separate review was another model context, not external peer review. No proof-assistant formalization is claimed. The mathematical argument, including the finite exclusions, is given in full above so that it can be checked independently of that assistance.

References

- [1] Zhichao Chen and Yimin Huang, *Mutation-preserving generalized cluster algebras and Laurent mutation invariants*, arXiv:2608.10551v3 (2026). <https://arxiv.org/abs/2608.10551v3>.
- [2] Yasuaki Gyoda and Kodai Matsushita, Generalization of Markov Diophantine equation via generalized cluster algebra, *Electronic Journal of Combinatorics* **30**(4) (2023), P4.10. <https://doi.org/10.37236/11420>.
- [3] Zijian Zeng, *Regularity Obstructions and a Certified Search for Rational Diophantine Septuples*, Zenodo preprint (August 2026). <https://doi.org/10.5281/zenodo.22055460>.